

Name: _____

SOLUTIONS

Date: _____



Year 10 Mathematics

Investigation 2, 2015

Topic – Linear Functions

In class component

$\frac{\quad}{56}$ $= \frac{\quad}{\quad} \%$

Total Time: **60** minutes

Weighting: _____ % of the year.

Equipment: Scientific Calculator, Take Home Component

AIM: In this assessment, you will be investigating parallel and perpendicular linear functions.

Take Home Component: 10 marks

1. [2 marks]

Write two rules about the relationship between the equations or graphs and the gradients of linear functions.
Hint: Refer to your take home component.

If the gradients are the same – linear functions are parallel ✓

If the gradients are reciprocals and opposite signs – linear functions are perpendicular ✓
i.e. gradients multiplied together equal -1.

2. [4 marks: 2, 2]

a) Which of the following are parallel to $y = 3x + 8$? Circle the correct answer(s).

$y = -3x + 8$	$y = 3x - 3$ ✓
$2y = 6x + 5$ ✓	$x = 3y + 8$

b) Which of the following are perpendicular to $y = 4x + 2$? Circle the correct answer(s).

$y = -\frac{1}{4}x + 3$ ✓	$4y = -x - 2$ ✓
$y = \frac{1}{4}x + 1$	$y = -4x + 6$

3. [18 marks: 8, 8, 1, 1]

a) For each equation given below, complete the table of values.

i) $y = 4x$

x	-2	-1	0	1	2
y	-8	-4	0	4	8

✓✓

ii) $y = 4x - 3$

x	-2	-1	0	1	2
y	-11	-7	-3	1	5

✓✓

iii) $y = -\frac{1}{4}x - 1$

x	-4	-2	0	2	4
y	0	-0.5	-1	-1.5	-2

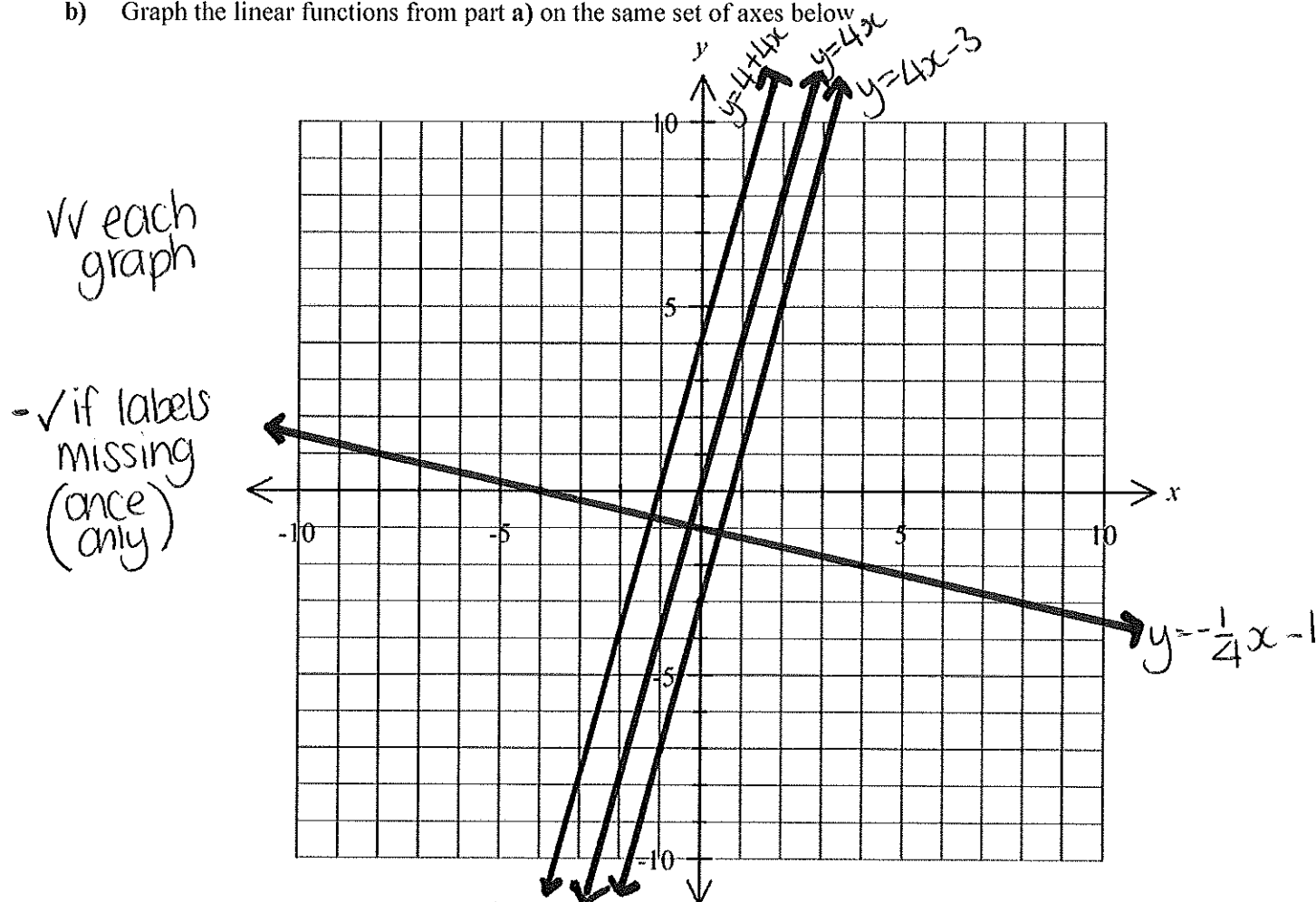
✓✓

iv) $y = 4 + 4x$

x	-2	-1	0	1	2
y	-4	0	4	8	12

✓✓

b) Graph the linear functions from part a) on the same set of axes below



c) Which lines are parallel?

$y = 4x$, $y = 4x - 3$, $y = 4 + 4x$ ✓

-1/2 each error or missing

d) Name one pair of lines that are parallel.

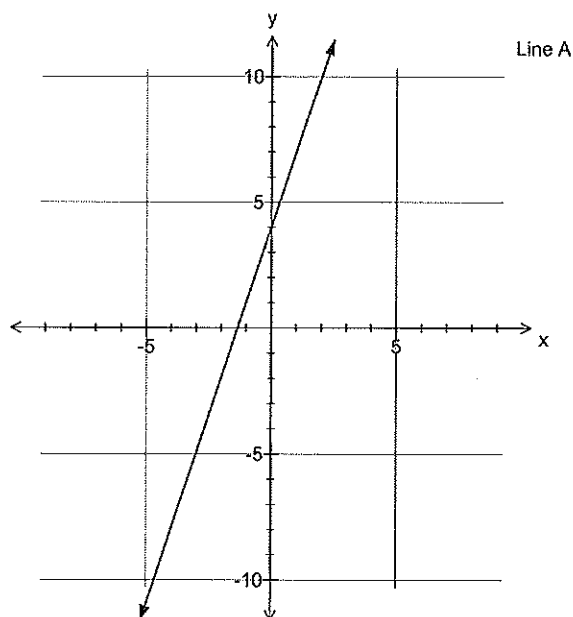
$y = -\frac{1}{4}x - 1$ and one other line.

✓✓

✓✓

4. [8 marks: 4, 4]

a)



What is the equation of line A?

$$m = \frac{4}{1} = 4$$

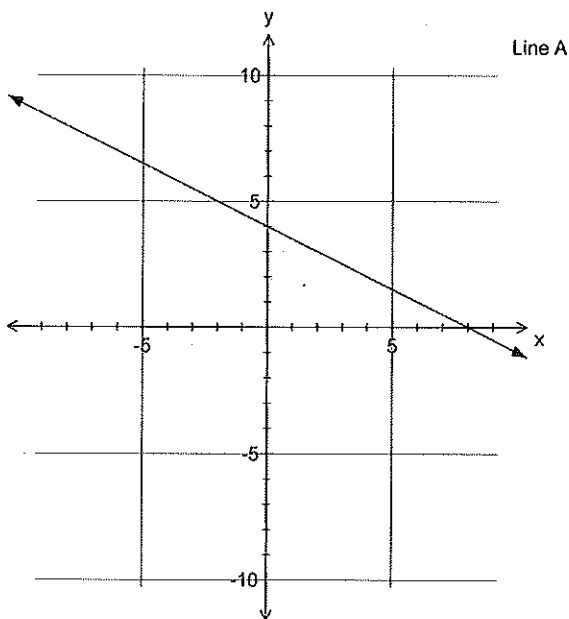
$$y = 4x + 4$$

Write the equation of a line which is parallel to line A.

$$y = 4x + c$$

must be 4 any value

b)



What is the equation of line A?

$$m = \frac{0 - 4}{8 - 0} = -\frac{1}{2}$$

$$y = -\frac{1}{2}x + 4$$

Write the equation of a line which is perpendicular to line A.

$$y = \frac{1}{2}x + c$$

must be $\frac{1}{2}$ any value

5. [4 marks: 2, 2]

a) What is the equation of the line that is parallel to $y = -5x + 6$ and crosses the y axis at (0,4)

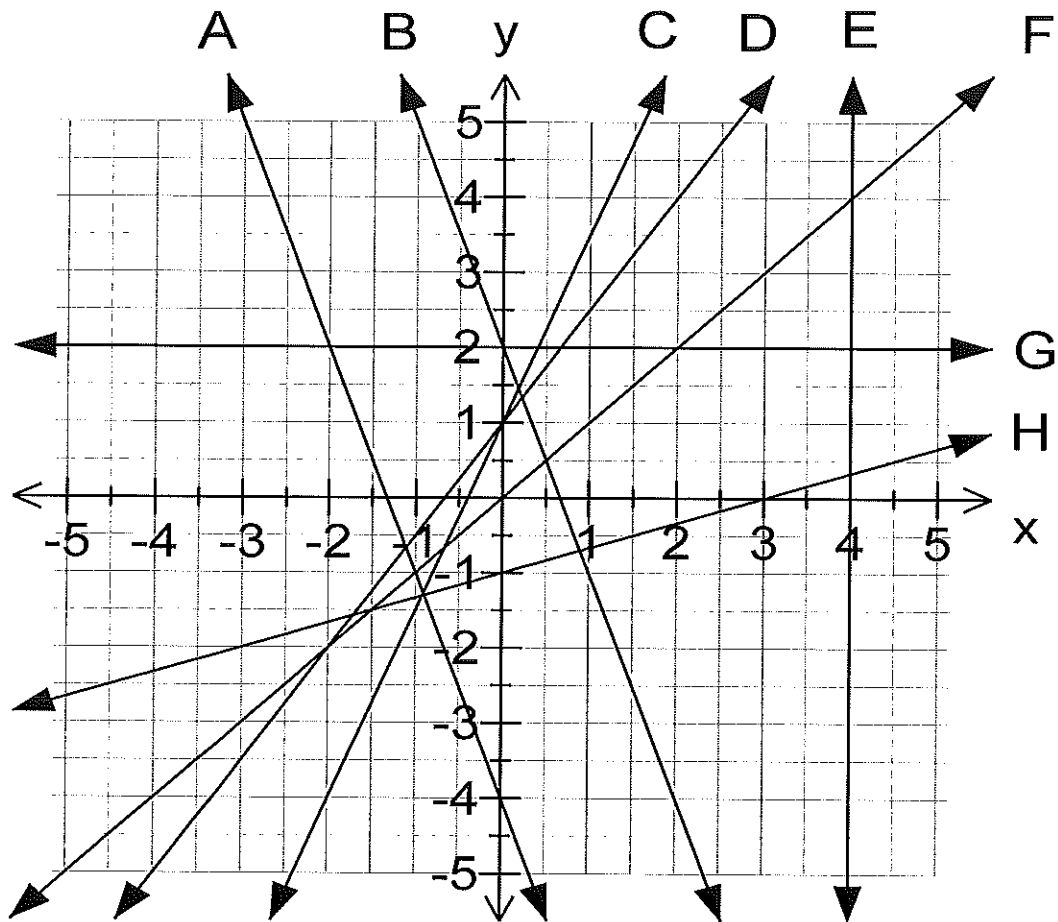
$$y = -5x + 4$$

b) What is the equation of the line that is perpendicular to $y = -\frac{1}{5}x + 3$ and crosses the y axis at (0,4).

$$y = 5x + 4$$

6. [10 marks]

The diagram below shows 8 lines, labelled A to H.



Determine which of these lines satisfy the following properties. In some cases, there will be more than one answer; include all cases.

a) Line(s) which are parallel	$A \nparallel B$	✓
b) Line(s) which have a positive gradient	C, D, F, H	✓
c) Line(s) which pass through the origin	F	✓
d) Line(s) with an x-intercept of 3	H	✓
e) Line(s) with a y-intercept of 1	C & D	✓
f) Line(s) which intersect at $(-2, 2)$	D & F	✓
g) Line(s) which are perpendicular	$G \perp H, A \perp H, B \perp H$	✓✓
h) Vertical lines	E	✓
i) Line(s) with equation $3x - 2y = -2$	D	✓

$$\begin{aligned}
 -2y &= -3x - 2 \\
 y &= 1.5x + 1
 \end{aligned}$$